

軟體測試與資安檢測

教師：王凡

授課時間：每週三 9：10—12：00

博理館 113

課程代號：EE 5170

識別碼：921 U9550

民國 110 年春季班

QUIZ (2021/3/24)

Open-book, open-internet, no discussion

Name: _____ Department : _____ Student ID: _____

1. Please draw the control-flow graph (CFG) of the following program code.

```
x = y
while x < 100:
    if x < y:
        x += 1
        break
    for z in (1, x):
        x = x + z
    if x > 5:
        y += 1
    else:
        y += 2
print (x, y)
```

2. For each node n in the control flow graph in problem 1, please write down $\text{def}(n)$ and $\text{use}(n)$.

3. Continuing from problem 1, for each variable a , please write down the DU pairs of a .

- Open-book! Open-internet! No discussion!.

6. Continuing from problem 4, please write down a test set that satisfies all-def coverage.
7. Continuing from problem 4, please write down a test set that satisfies all-use coverage.
8. Continuing from problem 4, please write down a test set that satisfies all-du-paths coverage.